Recurrence solutions

1

The fixed point solves $a^* = 2a^* - 1$, giving $a^* = 1$. Then $b_n=a_n-1$ has $b_n=2b_{n-1}$,$b_0=2$, hence $a_n=2^{n+1}+1$. Iteration gives $a_n = 2^n a_0 - \sum_{k=0}^{n-1} 2^k = 3 \cdot 2^n - (2^n - 1) = 2^{n+1} + 1$. The initial value is three, and $2a_{n-1}-1=2^{n+1}+1$ verifies substitution.

2

The polynomial $r^2-5r+6=(r-2)(r-3)$ gives $a_n=C_1 2^n+C_2 3^n$. Initial equations $C_1+C_2=1$,$2C_1+3C_2=4$ give $C_1=-1$,$C_2=2$, so $a_n=2 \cdot 3^n-2^n$.

3

The polynomial is $(r-2)^2$, giving $a_n=(C_1+C_2 n)2^n$. At zero $C_1=2$; at one $2(C_1+C_2)=6$ gives $C_2=1$. Thus $a_n=(n+2)2^n$.

4

The homogeneous root is two, giving $C2^n$. Forcing base three does not collide. Trial $A3^n$ gives $A3^n-2A3^{n-1}=A3^{n-1}=3^n$, so $A=3$. The general solution is $a_n=C2^n+3^{n+1}$.

5

Near the base $f(2)=2f(1)+2=4$ and $f(4)=2f(2)+4=12$. Conjecture $f(2^m)=(m+1)2^m$. The base $m=0$ gives one. If it holds at m, $f(2^{m+1})=2(m+1)2^m+2^{m+1}=(m+2)2^{m+1}$, proving induction.

For general integers, first prove monotonicity. The base $f(2)>f(1)$ holds. For $n\geq 2$,

$$f(n+1) - f(n) = 2[f(\lfloor (n+1)/2 \rfloor) - f(\lfloor n/2 \rfloor)] + 1.$$

At $n=2k$ the bracket vanishes; at $n=2k+1$ it is $f(k+1)-f(k)$, nonnegative by strong induction. Thus f is nondecreasing. For $m=\lfloor \log_2 n \rfloor$,

$$(m+1)2^m \leq f(n) \leq (m+2)2^{m+1}.$$

For $n\geq 2$ use $m+1\geq \log_2 n$,$2^m\geq n/2$,$m+2\leq 3\log_2 n$,$2^{m+1}<2n$ to obtain

$$\tfrac{1}{2} n \log_2 n \leq f(n) \leq 6n \log_2 n.$$

Hence $\Theta(n \log n)$. Floors change the exact path; this bounding proof does not pretend every n is a power of two.

Recurrence solutions (continued)

6

With $x_0=0$ all terms stay zero. With $x_0=1$, initial terms are 1,2,6,42 and continue growing without a bounded solution. Nonlinearity rules out the constant-coefficient linear characteristic-polynomial method, without proving that no conceivable closed form exists. Since $x_{n+1} \geq x_n^2$ and $x_1=2$, induction gives $x_n \geq 2^{2^{n-1}}$ for $n \geq 1$, proving unbounded growth.

7

The homogeneous solution is $C4^n$. Trial $An+B$ gives $An+B=(4A+3)n+(4B-4A)$. Thus $A=4A+3$ gives $A=-1$, and $B=4B+4$ gives $B=-4/3$. Consequently $b_n=C4^n-n-4/3$. Initial $b_0=1$ gives $C=7/3$:

$$b_n = (7 \cdot 4^n - 3n - 4)/3.$$

Checks give $b_1=7=4 \cdot 1+3$ and $b_2=34=4 \cdot 7+6$.

8

The homogeneous polynomial $(r-2)^2$ has root two of multiplicity two. Neither $A2^n$ nor $An2^n$ survives as a particular trial; use $An^2 2^n$. Substitution gives

$$An^2 2^n - 4A(n-1)^2 2^{n-1} + 4A(n-2)^2 2^{n-2} = 2A2^n.$$

Thus $A=3/2$ and $a_n=(C_1+C_2 n+3n^2/2)2^n$. Initial $a_0=1$ gives $C_1=1$; $2(1+C_2+3/2)=5$ gives $C_2=0$. Hence $a_n=(1+3n^2/2)2^n$. At $n=2$ it is $28=4 \cdot 5-4 \cdot 1+12$.

Alternatively, divide by 2^n and put $u_n = a_n/2^n$. Then $u_n - 2u_{n-1} + u_{n-2} = 3$, with $u_0 = 1, u_1 = 5/2$. Set $d_n = u_n - u_{n-1}$. Since $d_1 = 3/2$ and $d_n = d_{n-1} + 3$, we get $d_n = 3n - 3/2$. Summing yields $u_n = 1 + 3n^2/2$, matching coefficient comparison.

n	0	1	2	3	4
a_n	1	5	28	116	400
u_n	1	5/2	7	29/2	25

Checks give $a_3 = 4 \cdot 28 - 4 \cdot 5 + 3 \cdot 8 = 116$ and $a_4 = 4 \cdot 116 - 4 \cdot 28 + 3 \cdot 16 = 400$. The factor n^2 escapes the homogeneous solution space; the constants C_1, C_2 then fit the initial data of the total solution.

9

The roots of $(r-1)(r-3)$ give $a_n=C_1+C_2 3^n$. Equations $C_1+C_2=2, C_1+3C_2=7$ give $C_2=5/2, C_1=-1/2$, so $a_n=(5 \cdot 3^n-1)/2$.

Recurrence solutions (continued)

The OGF equations retain initial corrections:

$$(A-2-7x)-4x(A-2)+3x^2A=0, \qquad A=\frac{2-x}{(1-x)(1-3x)}.$$

Writing A=α/(1−x)+β/(1−3x), numerator comparison gives α+β=2,−3α−β=−1, hence α=−1/2,β=5/2. Extracting coefficients gives the same formula. A(0)=2 and A′(0)=7 verify the initial numerator.

10

Root splitting gives $C_0=1$ and the stated convolution. Summing $n\geq 1$ gives $C-1=xC^2$. The branch with C(0)=1 is $C=(1-\sqrt{1-4x})/(2x)$. The generalized binomial expansion of the square root, reindexed by one, gives $C_n=\binom{2n}{n}/(n+1)$. Initial values are 1,1,2,5, and $C_4=\binom{8}{4}/5=14$.

Legal push/pop histories give outputs 123,132,213,231,321. For 231, push 1, push 2, pop 2, push 3, pop 3, pop 1. To output 3 first, both 1 and 2 must already be below it, with 2 above 1. Then 2 must precede 1, ruling out 312. The five outputs match $C_3=5$.

11

$P(r)=r^3-2r^2=r^2(r-2)$, so zero has multiplicity two. A complete basis is $\delta_0,\delta_1,2^n$, where $\delta_j(n)=1$ iff $n=j$. Write $a_n=A\delta_0(n)+B\delta_1(n)+C2^n$. From $a_2=5$, C=5/4; from $a_0=3$, A=7/4; from $a_1=4$, B=3/2. Thus

$$a_n=\tfrac{7}{4}\delta_0(n)+\tfrac{3}{2}\delta_1(n)+\tfrac{5}{4}\,2^n.$$

The terms are 3,4,5,10,20,40; the $n\geq 2$ tail is $5\cdot 2^{n-2}$. Both 0^n and $n0^n$ vanish for $n\geq 1$, regardless of the index-zero convention, so cannot represent arbitrary a_1.

12

Internal levels j=0,1,2,3 have 1,2,4,8 nodes. Their total costs are 4,16/3,8,16; sixteen leaves cost one each. Hence T(16)=4+16/3+8+16+16=148/3.

Generally $n=2^k$ and level j costs n/(k−j), j=0,…,k−1. Including leaves gives $T(n)=n(1+H_k)$. Integral bounds $\ln(k+1)\leq H_k\leq 1+\ln k$ give $\Theta(n\log k)=\Theta(n\log\log n)$. At n=1 use the separate base rather than dividing by zero. The ratio $(n/\log n)/n$ tends to zero, so forcing is not $\Theta(n)$; it also has no fixed polynomial gap from critical n. The polynomial-cost theorem does not apply, while layer sums reveal harmonic accumulation.

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Recurrence solutions (continued)

13

Roots of r^2-r+1 are $(1\pm i\sqrt{3})/2=e^{\pm i\pi/3}$. The complex general solution is $Ae^{in\pi/3}+Be^{-in\pi/3}$; the real form is $C\cos(n\pi/3)+D\sin(n\pi/3)$. Initial values give $C=1$ and $1/2+D\sqrt{3}/2=0$, so

$$a_n=\cos(n\pi/3)-\sin(n\pi/3)/\sqrt{3}.$$

The first seven values are $1,0,-1,-1,0,1,1$. Both trigonometric terms are unchanged by $n\mapsto n+6$, proving period six. Alternatively $(a_6,a_7)=(1,0)$ returns to the initial pair and recurrence uniqueness proves repetition. A finite observed list alone would not prove it.

14

Summing from $n\geq 1$ gives $A-1=2xA+xB, B=xA+xB$. Thus $B=xA/(1-x)$, and

$$[(1-2x)(1-x)-x^2]A=1-x,\quad A=\frac{1-x}{1-3x+x^2},\quad B=\frac{x}{1-3x+x^2}.$$

For $r_\pm=(3\pm\sqrt{5})/2$, their sum is three and product one. Numerator checks give

$$B=\frac{1}{\sqrt{5}}\left(\frac{1}{1-r_+x}-\frac{1}{1-r_-x}\right),\quad A=\frac{1}{\sqrt{5}}\left(\frac{r_+-1}{1-r_+x}+\frac{1-r_-}{1-r_-x}\right).$$

Hence

$$b_n=(r_+^n-r_-^n)/\sqrt{5},\quad a_n=((r_+-1)r_+^n+(1-r_-)r_-^n)/\sqrt{5}.$$

Direct pairs are $(1,0),(2,1),(5,3),(13,8),(34,21)$. Each component satisfies $u_n=3u_{n-1}-u_{n-2}$ with initial pairs $(1,2)$ and $(0,1)$. In particular $a_4=3\cdot 13-5=34, b_4=3\cdot 8-3=21$, agreeing with simultaneous updates.

Generally $G-y_0=xMG$. The series $H=\sum_{k\geq 0}x^kM^k$ has a finite matrix at every coefficient and satisfies $(I-xM)H=H(I-xM)=I$ by cancellation M^k-M^k at every positive degree. Therefore $G=Hy_0=(I-xM)^{-1}y_0$. Its inverse exists because the constant matrix of $I-xM$ is I, without requiring M invertible. For $M=0$, $G=y_0$ still agrees.

References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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